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On compacton traveling wave solutions of Zakharov-Kuznetsov-Benjamin-Bona-Mahony (ZK-BBM) equation

Nadeem Salamat[1] · Ameer Hamza Arif[1] · Muhammad Mustahsan[2] · Malik M. Saad Missen[3] · V. B. Surya Prasath[4]

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Abstract
Nonlinear evolution equations are used to model various complex physical phenomena, and finding exact solutions of these equations is important for their applications. One of the important problem is that of finding traveling wave solutions in well-known nonlinear evolution systems from mathematical physics. In this paper, the differential transformation method (DTM) along with the wave-like transformation is used to develop the compacton traveling wave solution of nonlinear Zakharov-Kuznetsov-Benjamin-Bona-Mahony (ZK-BBM) equation. The multiple choices for the parameters in the partial differential equation are studied and results shows that the method is efficient to develop new compacton wave solutions of partial differential equation. We plot the exact analytical and numerical solutions for different values of parameters. These results validate the experiment results.

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✉ Nadeem Salamat
nadeem.salamat@kfueit.edu.pk

Ameer Hamza Arif
hamza8644@gmail.com

Muhammad Mustahsan
muhammad.mustahsan@iub.edu.pk

Malik M. Saad Missen
saad.missen@iub.edu.pk

V. B. Surya Prasath
prasatsa@uc.edu

1 Department of Mathematics, Khawaja Fareed University of Engineering and Information Technology, Rahim Yar Khan, Pakistan

2 Department of Mathematics, The Islamia University of Bahawalpur, Bahawalpur, Pakistan

3 Department of Computer Science and IT, The Islamia University of Bahawalpur, Bahawalpur, Pakistan

4 Department of Electrical Engineering and Computer Science, University of Cincinnati, Cincinnati, OH 45221, USA

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1 Introduction

The nonlinear partial differential equations are used in modeling complex real-world phenomenon such as wave propagation, heat and fluid flow. The solution of nonlinear partial differential equations, which are not directly integrable, represents a special phenomenon of the medium. This phenomenon is known as cancelation effect of nonlinearity and dispersive effects in the medium and it is represented by many types of traveling waves such as soliton, cnoidal, kink, compacton and peakon solutions (Rawashdeh and Maitama 2017; Yao et al. 2017; Wazwaz 2007; Zhang et al. 2020). The example of such solutions is compacton wave solutions (Sergyeyev et al. 2019; Bazeia et al. 2010; Zhang and Huang 2019; Zilburg and Rosenau 2019). However, it is typical to find such solutions, as a few work is cited in literature and Zakharov-Kuznetsov-Benjamin-Bona-Mahony (ZK-BBM) equation is one of them. The water wave with smaller amplitude and larger wavelength are studied by ZK-BBM equation which is introduced by combining the Benjamin-Bona-Mahony (BBM) and Zakharov-Kuznetsov (Zk) equation (Galdi and Zhao 2014). This equation represents the unidirectional propagation of long waves in certain nonlinear dispersive systems. Its solition solutions are developed using the Adomian's decomposition (Dhaigude et al. 2012), approximate analytical solution (Tasbozan and Kurt 2015), $\frac{G'}{G}$ method (Shakeel and Mohyud-Din 2015), exp-function method (Ali et al. 2016), variational iteration methods (VIM) (Mohyud-Din et al. 2010; Patel and Kumar 2018), Homotopy Perturbation Method (HPM) (Mirgolbabaei et al. 2009), and sine-cosine method (Bibi and Mohyud-Din 2014). The cubic B spline method (Jiwari et al. 2019; Kumar et al. 2021; Jiwari et al. 2022; Pandit et al. 2017)

The differential transformation method (DTM) was introduced in Zhou (1986). This method is used to determine the numerical solution of many types of ODEs, PDEs and traveling wave solutions of PDEs. In this paper, we adapted the DTM to solve ZK-BBM equation to find the solitary wave solution. Our experimental results with different parameter sets of ZK-BBM indicate the behavior of solutions.

Rest of this paper is structured as follows. In Sect. 2, we briefly describe the DTM method, and the ZK-BBM Equation, its analytical wave-like solutions are discussed. Numerical solutions of ZKBBM equation with DTM for various parameter options are given in Sect. 3. We also present graphical profiles of the numerical solutions obtained. Sect. 4 concludes the work.

2 ZK-BBM Equation

2.1 Description of the DTM Method

The differential transformation method (DTM) introduced by Zhou (1986) establishes the numerical solution of differential equations. Nowadays, this method is used for solving linear and nonlinear differential equations numerically (Mirzaee 2011; Jafari et al. 2010; Yigider

and Karabacak 2015; Abdel-Halim 2008; Ravi Kanth and Aruna 2009). This method is also used for determining the singular value solutions (Ertürk 2007) and system of differential equations (Hussin and Kilicman 2013; Munganga et al. 2014; Hussin et al. 2010; Ayaz 2003). An extension of the DTM, called Fuzzy Differential Transformation Method (FDTM) is proposed in Allahviranloo et al. (2011), which is implemented to find the solutions of Fuzzy Differential Equations (FDE) and Fuzzy Initial Value Problems (FIVP) (Allahviranloo et al. 2009; Ahmadi and Kiania 2013) and fuzzy integral equations (Salahshour and Allahviranloo 2013; Odibat 2008; Hadizadeh and Moatamedi 2007). The fundamental theorems of DTM are represented as follows.

Definition 1 The differential transformation of the function $f(x)$ for the kth derivative is defined as follows (Faghanpour et al. 2016)

$$U(k) = \frac{1}{k!}\left(\frac{d^k f(x)}{dx^k}\right)_{x=x_0} \tag{1}$$

Where $f(x)$ is the original function , k is a positive integer and $U(k)$ is the transformed function.

Definition 2 The inverse transformation of $U(k)$ is defined as the polynomial sum of the kth order series, see (Zhou 1986).

$$f(x) = \sum_{k=0}^{\infty}(x - x_0)^k U(k) \tag{2}$$

By combining Eqs. (1) and (2) we get from

$$f(x) = \sum_{k=0}^{\infty}(x - x_0)^k \frac{1}{k!}\left(\frac{d^k f(x)}{dx^k}\right)_{x=x_0} \tag{3}$$

This fundamental definitions are used to derive the following theorem.

Theorem 1 *Let $f(x)$ and $g(x)$ are two functions and $F(k)$ and $G(k)$ are the differential transformation of $f(x)$ and $g(x)$, respectively, and α and β are constant. All the theorems are derived using Eqs. 1 and 2), further details are given in Erturk and Momani (2007).*

a) If $u(x) = f(x) \pm g(x)$ then $U(k) = F(k) \pm F(k)$
b) If $u(x) = \alpha f(x)$ then $U(k) = \alpha F(k)$
c) If If $u(x) = \alpha f(x) \pm \beta g(x)$ then $U(k) = \alpha F(k) \pm \beta G(k)$
d) If $u(x) = \frac{d^n f_1(x)}{dx^n}$ then $U(k) = \frac{(k+n)!}{k!} F_1(k+n)$.
e) If $u(x) = f(x)g(x)$ then $U(k) = \sum_{k_1=0}^{k} F(k_1)G(k-k_1)$
f) If $u(x) = e^{\lambda x}$ then $U(k) = \frac{\lambda^k}{k!}$, where λ is a constant.
g) If $u(x) = x^m$ then $U(k) = \delta(k-m)$ where

$$\delta(k-m) = \begin{cases} 1, & if\, k = m, \\ 0, & if\, k \neq m. \end{cases}$$

h) If $u(x) = \sin(\alpha x + \beta)$ then $U(k) = \frac{\alpha^k}{k!}\sin\left(\frac{k\pi}{2} + \beta\right)$
i) If $u(x) = \cos(\alpha x + \beta)$ then $U(k) = \frac{\alpha^k}{k!}\cos\left(\frac{k\pi}{2} + \beta\right)$

This method is used for solving PDEs in two ways. Firstly, this method is extended for more than one variables and use directly to solve PDEs. Secondly, the wave-like transformation is used and target PDE is converted to the ODEs. At next step, simple DTM is used for developing solution of resultant ODE. We use the second approach to solve the nonlinear ZK-BBM equation.

2.2 The ZK-BBM equation

The nonlinear ZK-BBM equation represents the water wave phenomena of longer wavelength and smaller amplitude. It is a regularized long wave equation, 2D case is represented as:

$$u_t + u_x - 2auu_x - bu_{xxt} = 0 \qquad t > 0 \tag{4}$$

The water wave under the consideration of smaller amplitude and larger wavelength are studied by ZK-BBM equation. This represents the unidirectional propagation of long waves in certain nonlinear dispersive systems and has been introduced in order to give us a observation for the smaller wave amplitude $u(x, t)$ in the large wavelength in many physical model. Thus waves of extreme length, along the $+x$ propagation direction, are described by $u_t + c_0 u_x = 0$ whose constant solution is an arbitrary differentiable function of $x \pm c_0 t$. We used this wave-like transformation with differential transformation method to solve ZK-BBM equation.

2.3 Solitary wave solution of Zk-BBM Equation

Consider ZK-BBM equation in $2D$ case as Eq. (4). Let the Eq. (4) has solution $u(x, t)$. First of all, we invoke the wave-like transformation $\eta = kx + \omega t$ where k is the $+ve$ constant and represents the wave number and ω is the wave speed at time t. This choice of transform changes Eq. (4) into a simple ordinary differential equation of one variable. As a result, Eq. (4) becomes

$$(\omega + k)u' - 2akuu' - b\omega k^2 u''' = 0 \tag{5}$$

where a, b and ω are arbitrary constants and wave phase, respectively, and u', u'', u''' represents the ordinary derivatives with respect to the newly introduced variable η. First of all, we solve this equation analytically with the conditions of solitary wave solution.

For exact solution, integrate Eq. (5) with respect to η and constants are determined by applying the initial conditions derived from the data. The conditions for solitary wave solution are $\eta \to \pm\infty$ then $u, u', u''.... \to 0$. Then we have a solitary wave.

$$u(\eta) = \frac{K'}{B} \sec h^2\left(\frac{-\sqrt{K'}}{2}\eta\right) \tag{6}$$

where $K' = \frac{\omega+k}{b\omega k^2}$ and $B = \frac{2a}{3b\omega k}$. By backward substitution of wave-like transformation, we get the solution of Eq. (4) as

$$u(x, t) = \frac{K'}{2B} \sec h^2\left(\frac{\sqrt{K'}}{2}(kx + \omega t)\right) \tag{7}$$

this is the special case of d'Alembert's solution, where coordinates move along the wave $\eta = kx + ct$ and its shape remains similar unchanged as described by

$$u(x, t) = \frac{K'}{2B} \sec h^2\left(\frac{\sqrt{K'}}{2}(kx)\right)$$

at $t = 0$. But this wave is shifted to left by an amount of ct. These results are shown graphically for multiple values of k and ω in Fig.(1), with the first column shows the profile of the solution function with a fixed $t = 0$ and x for the mentioned domain, and second column shows the profile of solution function with range of a variables x, T.

The graphical representation shows that profile of the solution is bell shaped. The parameters k and ω does not change the profile of the wave-like solution. The change in ω effects only the wave phase.

3 Solution of ZK-BBM differential Equation by DTM

The DTM is a powerful tool for investigation of analytical numerical solutions of linear and nonlinear differential and partial differential equations and integral equations. The method is frequently used to analyze the numerical solutions of nonlinear problems arising in sciences. In this section we develop the numerical solution of ZK-BBM equation described in Eq. (4) by applying differential transformation. we applied the differential transformation as mentioned in theorem 1 in section (2.1) to Eq. (5). As a result, the following algebraic relation is obtained;

$$(\omega + k)(j+1)U(j+1) - 2ak\left(\sum_{r=0}^{j} U(r)U(j+1-r)(j+1-r)\right)$$
$$-b\omega k^2\Big((j+3)(j+2)(j+1)U(j+3)\Big) = 0 \tag{8}$$

where j is variable by rearranging terms. This equation is simplified for $U(j+3)$, as a result, we get,

$$U(j+3) = \frac{(\omega + k)(j+1)U(j+1) - 2ak\Big(\sum_{r=0}^{j} U(r)U(j+1-r)(j+1-r)\Big)}{b\omega k^2(j+3)(j+2)(j+1)}. \tag{9}$$

This relation is used to determine the coefficients of the series solution developed by the differential transformation method. Now we discuss the impact of constants a and b on the wave-like solution. These are summarized as a cases in next subsection, First of all, we generate the initial condition from the analytical solution for the predefined parameters $\omega = 1$ and $k = 1$. As the other cases are proportional to these solutions. The initial conditions become

$$u(0) = \frac{K'}{2B}; u'(0) = 0; u''(0) = -\frac{(K')^2}{4B} \tag{10}$$

where

$$K' = \frac{2}{b}; \qquad B = \frac{2a}{3b}, \tag{11}$$

then equation (9) becomes

$$U(j+3) = \frac{(2)(j+1)U(j+1) - 2a\Big(\sum_{r=0}^{j} U(r)U(j+1-r)(j+1-r)\Big)}{b(j+3)(j+2)(j+1)} \tag{12}$$

In what follows, we discuss in detail cases of the solutions using values of parameters a and b in intervals $(-\infty, 0]$ and $[0, \infty)$.

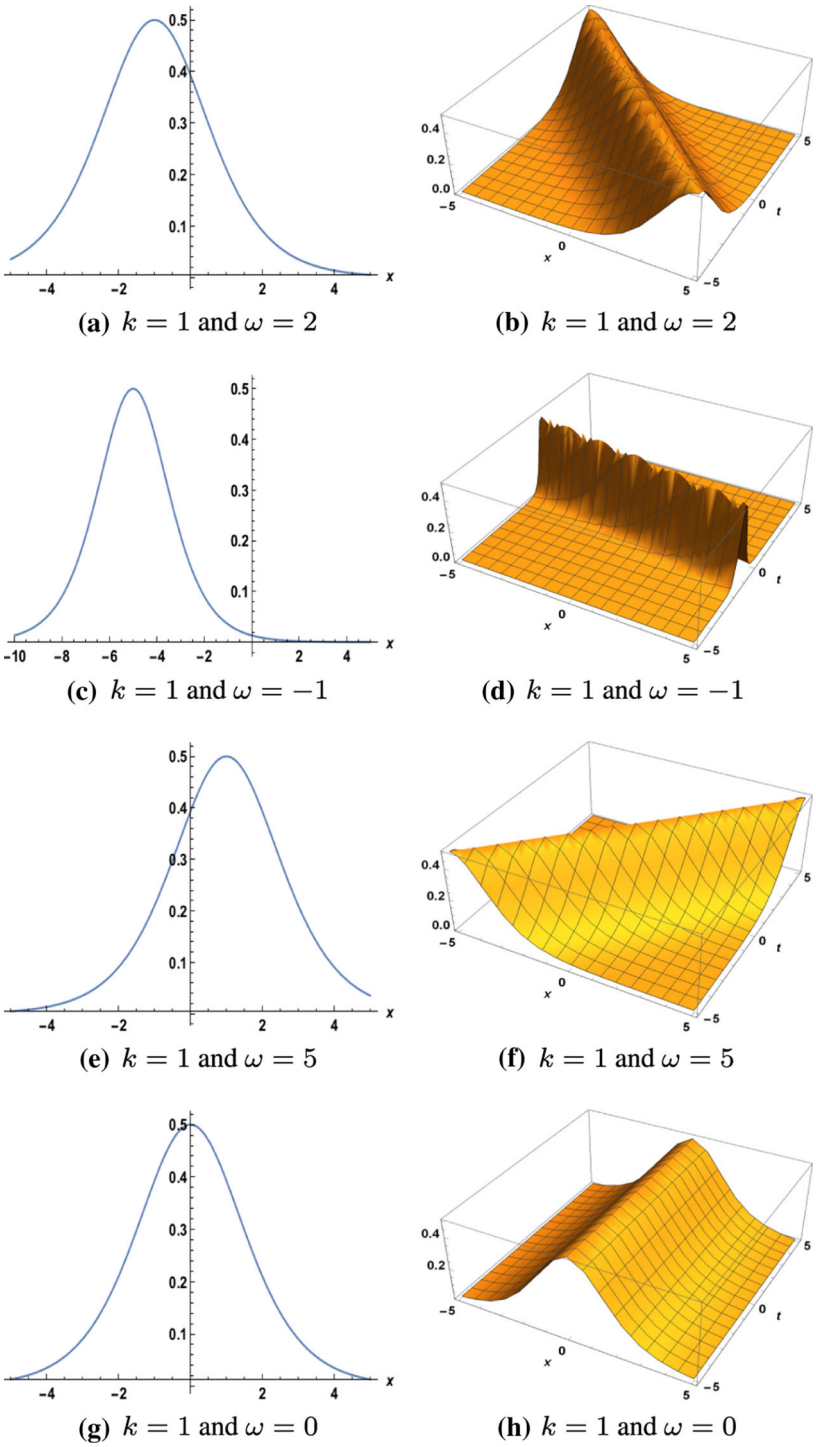


Fig. 1 The analytical solution of ZK-BBM equation represented graphically for multiple values of constant k and various wave phases ω

3.1 Case-I: $a > 0,\ b > 0$

In this case, we consider both a and b to be positive integers. For this purpose,let $a = m$ and $b = n$ where $m, n \in Z^+$, then Eq. (11) becomes,

$$K^{'} = \frac{2}{n};\ B = \frac{2m}{3n}$$

and initial conditions will be

$$u(0) = \frac{3}{2m} \qquad u^{'}(0) = 0 \qquad u^{''}(0) = -\frac{3}{2mn}$$

Apply DTM to transform initial condition we have $U(0) = \frac{3}{2m}$; $U(1) = 0$; $U(2) = -\frac{3}{4mn}$. Using these values of parameters, initial conditions in Eq. (12), we get

$$\begin{aligned} U(3) &= 0, & U(4) &= \frac{1}{16mn^2} \\ U(5) &= 0, & U(6) &= -\frac{1}{48mn^3} \\ &\ldots\ldots, & &\ldots,\ldots \\ &\ldots\ldots, & &\ldots,\ldots \end{aligned}$$

Now we used Eq. (2) and above coefficients, and backward substitution with particular choice of parameters in wave-like transformation, e.g., $\omega = 1$ and $k = 1$, we obtain the series solution of Eq. (4) and Eq. (2)

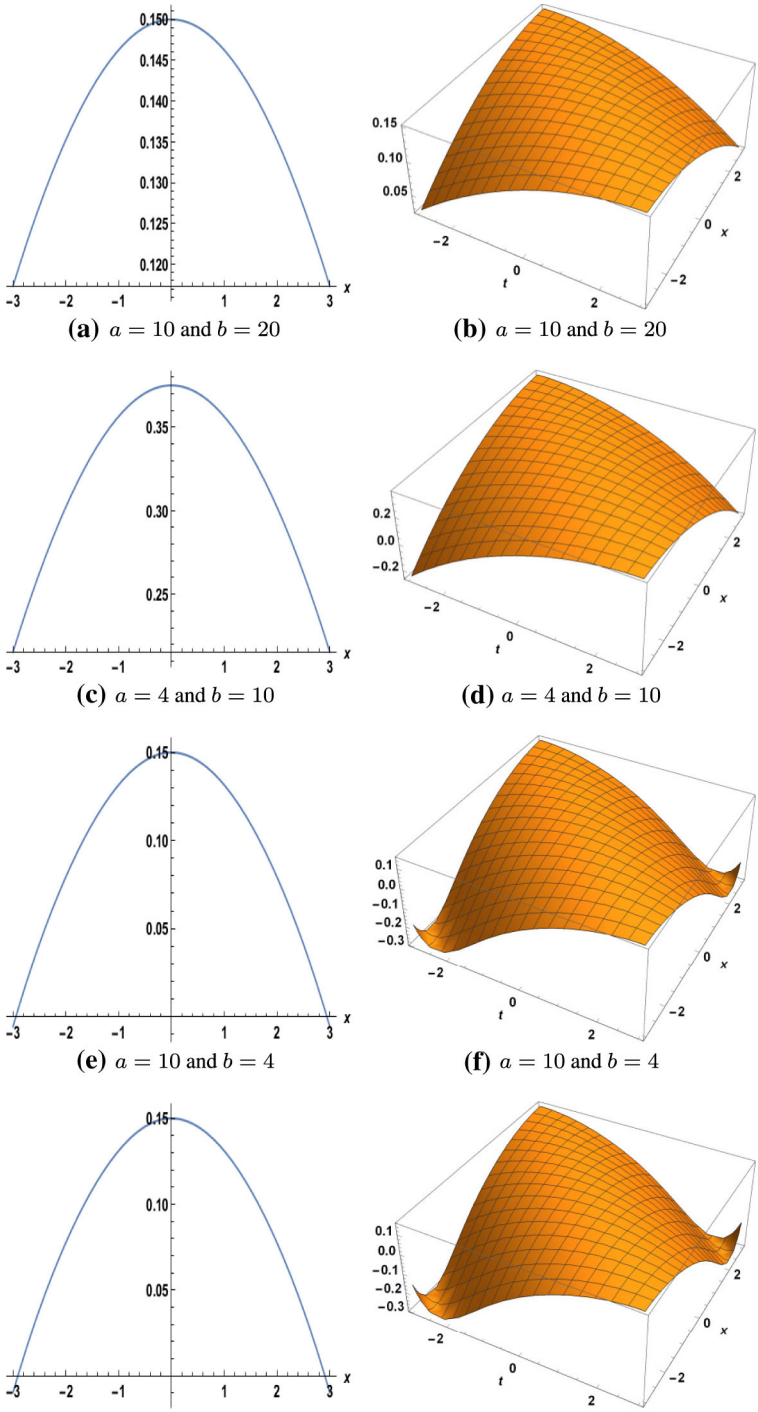
$$u_1(x+t) = \frac{3}{2m} - \frac{3}{4mn}(x+t)^2 + \frac{1}{16mn^2}(x+t)^4 - \frac{1}{48mn^3}(x+t)^6 + \frac{11}{5376mn^4}(x+t)^8 \ldots \quad (13)$$

The graphs of solution for the case are shown in Fig. 2 for $k = 1$ and $\omega = 1$ with multiple values of the parameters a and b. The analysis of the profile shows that parameters control the smoothness of the graph. In this figure, first column represents the wave-like profile of solution with $t = 0$ and in second column, the profile is drawn for a complete range of x and t. The profile shows that plateau of the campacton moves upward with the decrease in a, its top hill shrinks and wave top becomes sharper. At the mean time parameter b controls the wave length, wave length is directly proportional to b as visible in Fig. 2.

It is observed that parameter a does not effects much the shape of the wave, as parameter b controls. The wave top lowers down with the decrease in parameter b, at $b = 0.2$ it becomes flat and a new wave shape emerges when $b \in [0.20]$. This impact on wave is visible in different cases if Fig. 3.

3.2 Case-II $a < 0, b > 0$

For these choices of parameters, let $a = -m$ and $b = n$ where $m, n \in Z^+$. Put this values in Eq. (11) then we have $K^{'} = \frac{2}{n}$; $B = -\frac{2m}{3n}$ and initial conditions as $u(0) = -\frac{3}{2m} \quad u^{'}(0) = 0, \quad u^{''}(0) = \frac{3}{2mn}$



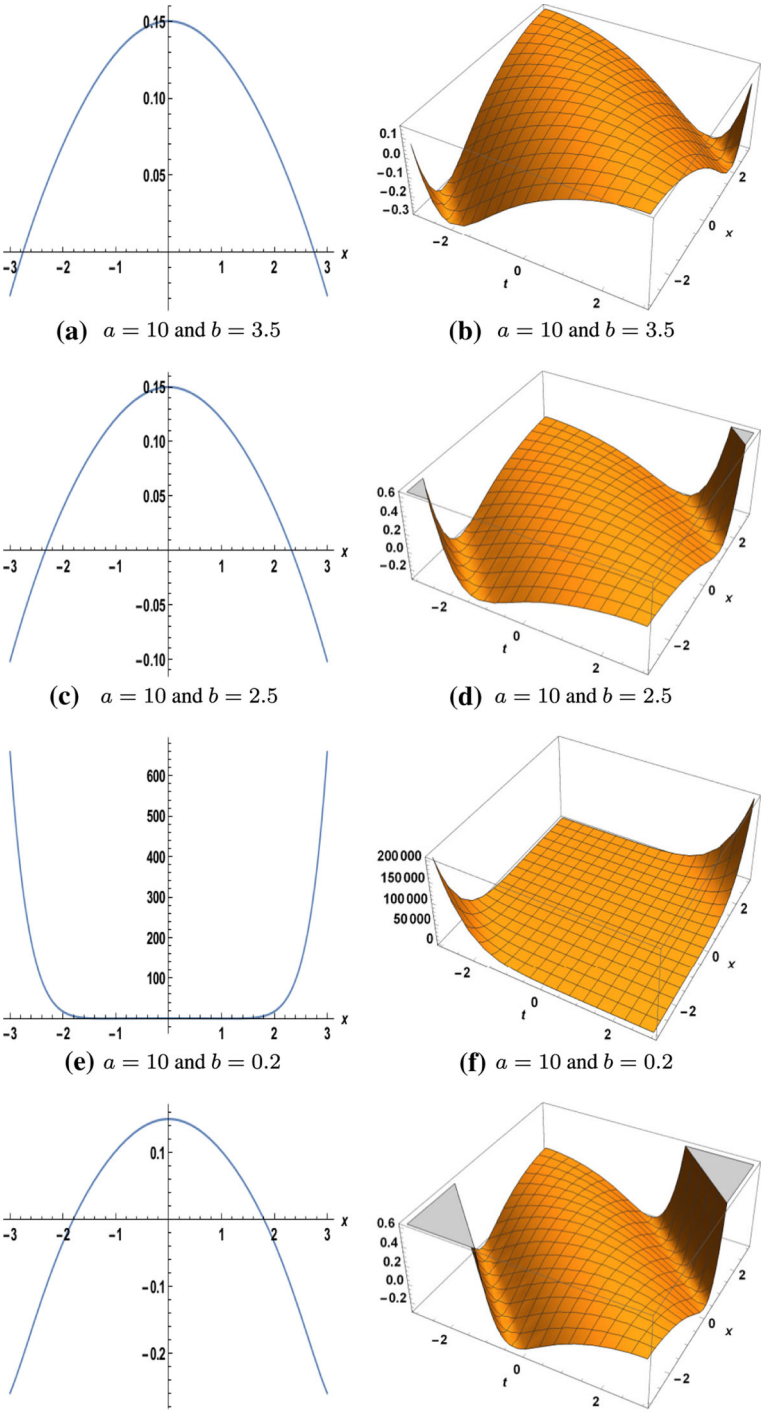
(a) $a = 10$ and $b = 20$ **(b)** $a = 10$ and $b = 20$

(c) $a = 4$ and $b = 10$ **(d)** $a = 4$ and $b = 10$

(e) $a = 10$ and $b = 4$ **(f)** $a = 10$ and $b = 4$

(g) $a = 10$ and $b = 3.9$ **(h)** $a = 10$ and $b = 3.9$

Fig. 2 Plots of the traveling wave solutions obtained by DTM against parameters a, b, both are positive



(a) $a = 10$ and $b = 3.5$ **(b)** $a = 10$ and $b = 3.5$

(c) $a = 10$ and $b = 2.5$ **(d)** $a = 10$ and $b = 2.5$

(e) $a = 10$ and $b = 0.2$ **(f)** $a = 10$ and $b = 0.2$

(g) $a = 10$ and $b = 1.5$ **(h)** $a = 10$ and $b = 1.5$

Fig. 3 Plots of the traveling wave solutions obtained by DTM against parameters a, b, both are positive

Apply DTM to initial condition we have $U(0) = -\frac{3}{2m}$; $U(1) = 0$; $U(2) = \frac{3}{4mn}$. Use all these values in Eq. (12), we get the differential transformation coefficients

$$\begin{aligned} U(0+3) &= 0 & U(4) &= -\frac{1}{16mn^2} \\ U(5) &= 0 & U(6) &= \frac{1}{48mn^3} \\ U(7) &= 0 & U(8) &= -\frac{1}{786mn^4} \\ &\cdots\cdots & &\cdots\cdots \\ &\cdots\cdots & &\cdots\cdots \end{aligned}$$

The Eq. (2) with above coefficients, and particular choice of $\omega = 1$ and $k = 1$ then $\eta = x + t$ equation Eq. (2) becomes

$$u_3(x+t) = -\frac{3}{2m} + \frac{3}{4mn}(x+t)^2 - \frac{1}{16mn^2}(x+t)^4 + \frac{1}{48mn^3}(x+t)^6 - \frac{11}{5376mn^4}(x+t)^8 \ldots \quad (14)$$

This shows that $u_2(x,t) = -u_1(x,t)$, i.e., Eqs. (14) and (13) are additive inverses of each other. Now we analyze the profile of the solution in Fig. 4. It is also obvious from Figs. 2, 3 and 4, that profile is also inverted.

It is also observed that the opposite effects occur in profile. In other fashion, it can be described as the compacton wave moves ahead or backward to the phenomenon discussed in case-I.

3.3 Case-III for $a < 0$, $b < 0$

Let $a = -m$ and $b = -n$ where $m, n \in Z^+$. Put this values Eq. (11), then we have $K' = -\frac{2}{n}$; 05cm$B = \frac{2m}{3n}$

$$u(0) = -\frac{3}{2m}, \qquad u'(0) = 0, \qquad u''(0) = -\frac{3}{2mn}$$

Applying differential transformation to initial conditions, then we have $U(0) = -\frac{3}{2m}$; $U(1) = 0$; $U(2) = -\frac{3}{4mn}$. From all the above conditions along with the Eq. (12), we get the differential transformation coefficients as

$$\begin{aligned} U(3) &= 0, & U(4) &= -\frac{1}{16mn^2} \\ U(5) &= 0, & U(6) &= -\frac{1}{48mn^3} \\ U(7) &= 0, & ,U(8) &= -\frac{1}{786mn^4} \\ &\cdots\cdots & &\ldots,\ldots,\ldots \\ &\cdots\cdots & &\ldots,\ldots,\ldots \end{aligned}$$

Put these values in Eq. (2), we get and for particular choice of parameters in transformation, such as $\omega = 1$ and $k = 1$ then $\eta = x + t$,

$$u_3(x+t) = -\frac{3}{2m} - \frac{3}{4mn}(x+t)^2 - \frac{1}{16mn^2}(x+t)^4 - \frac{1}{48mn^3}(x+t)^6 - \frac{1}{786mn^4}(x+t)^8 \ldots \quad (15)$$

The profile of this solution is represented in Fig. 5.

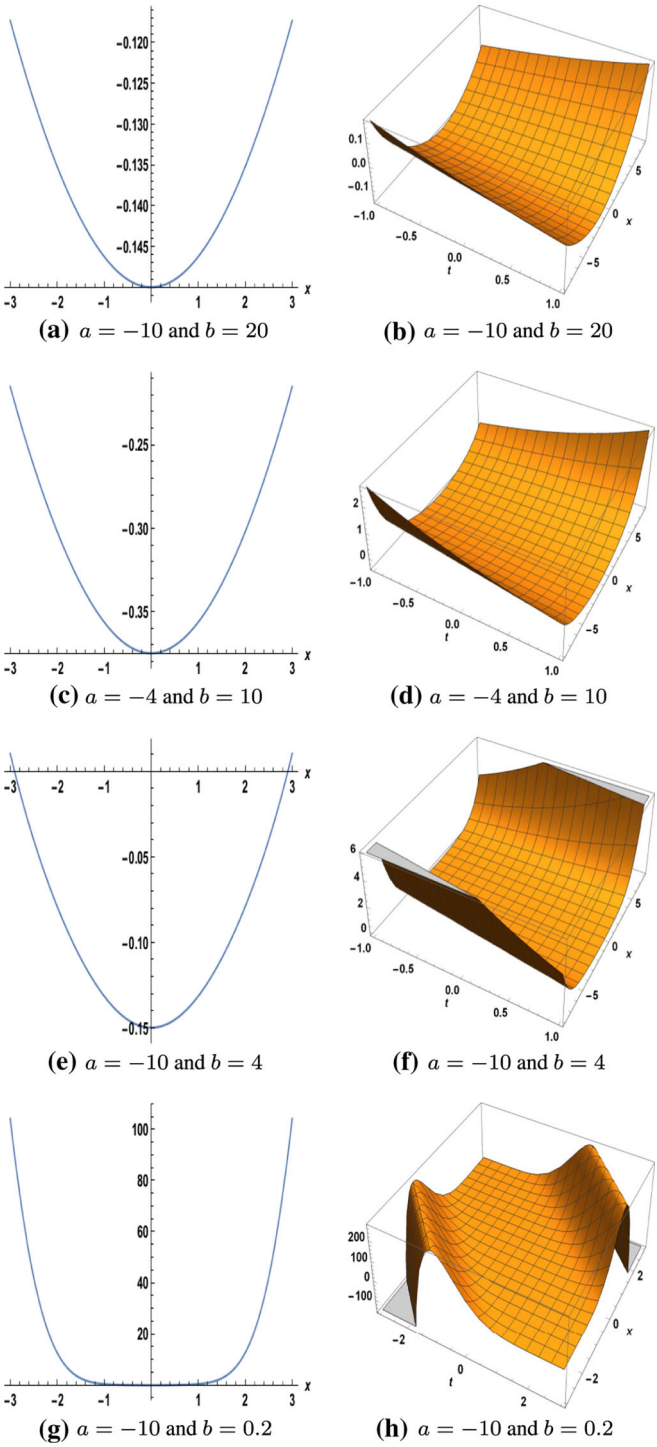
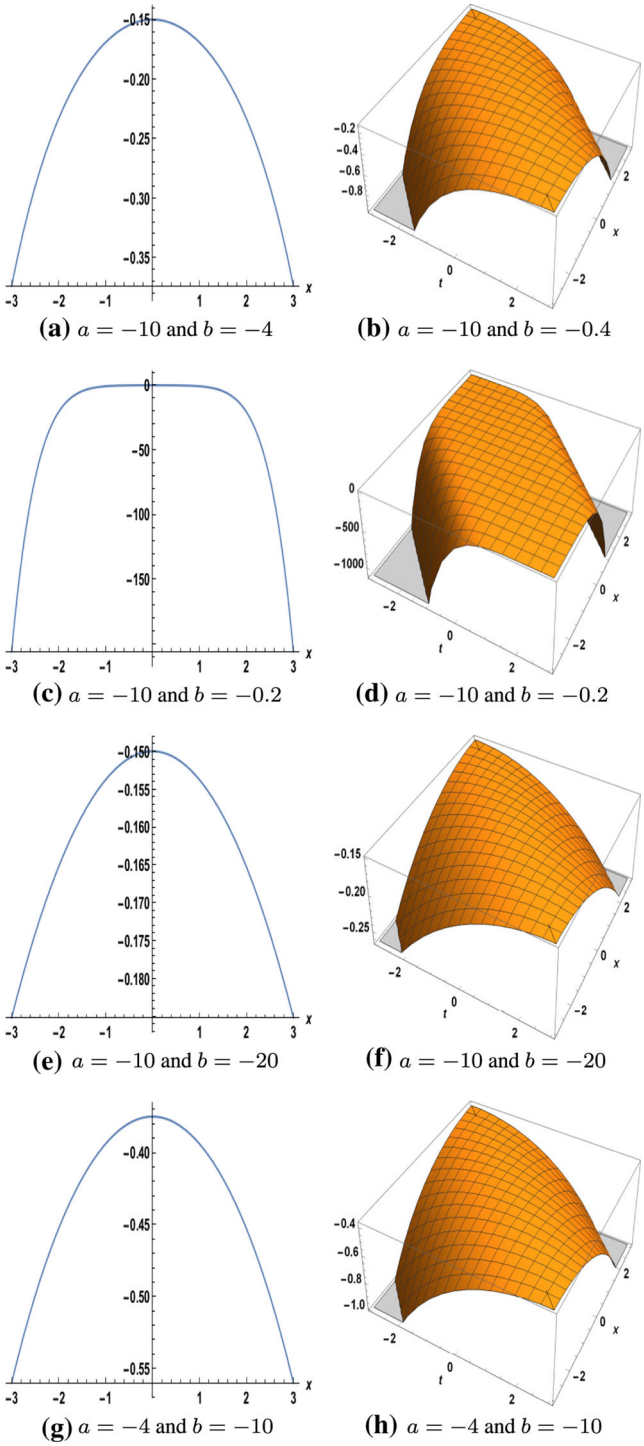


Fig. 4 Plots of the traveling wave solutions obtained by DTM against negative values of a and positive values of b



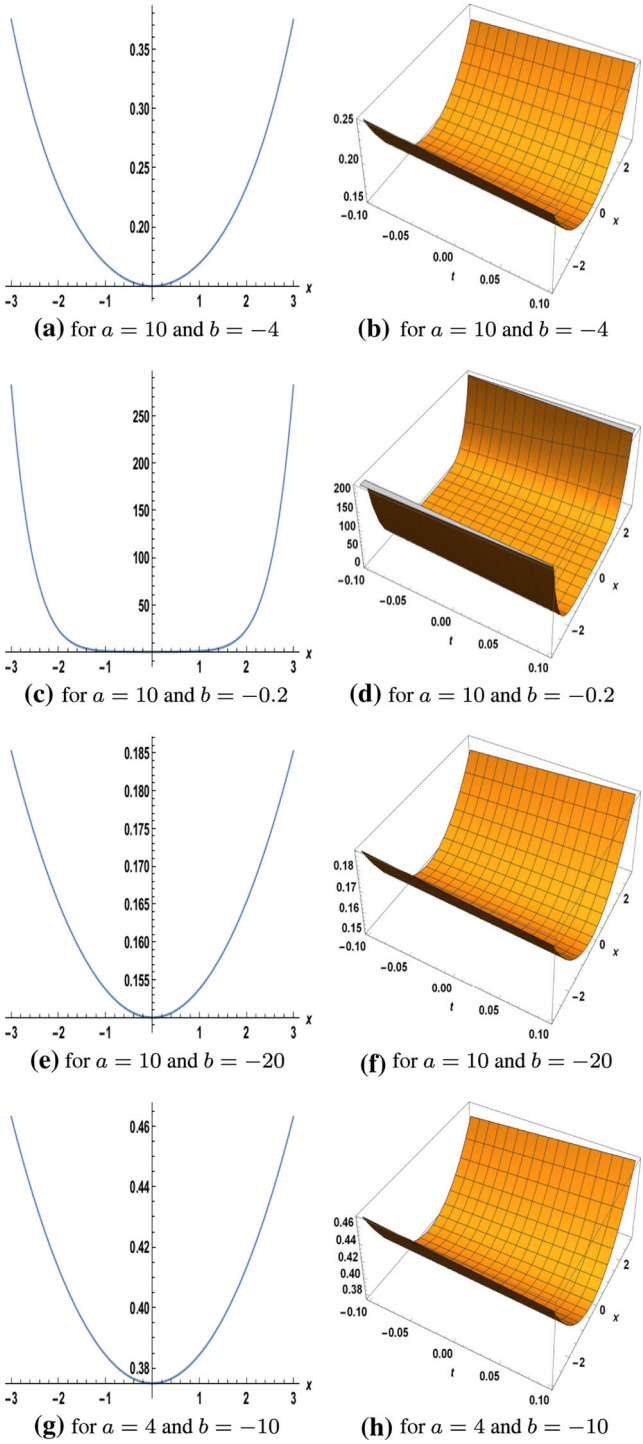
(a) $a = -10$ and $b = -4$ **(b)** $a = -10$ and $b = -0.4$

(c) $a = -10$ and $b = -0.2$ **(d)** $a = -10$ and $b = -0.2$

(e) $a = -10$ and $b = -20$ **(f)** $a = -10$ and $b = -20$

(g) $a = -4$ and $b = -10$ **(h)** $a = -4$ and $b = -10$

Fig. 5 Plots of the traveling wave solutions obtained by DTM against different negative values of a and b



(a) for $a = 10$ and $b = -4$ **(b)** for $a = 10$ and $b = -4$

(c) for $a = 10$ and $b = -0.2$ **(d)** for $a = 10$ and $b = -0.2$

(e) for $a = 10$ and $b = -20$ **(f)** for $a = 10$ and $b = -20$

(g) for $a = 4$ and $b = -10$ **(h)** for $a = 4$ and $b = -10$

Fig. 6 Plots of the traveling wave solutions obtained by DTM against positive values of a and negative values of b

3.4 Case-IV for $a > 0,\ b < 0$

For this choice of parameters, let $a = m$ and $b = -n$ where $m, n \in Z^+$. Put this values in the equation,

$$K^{'} = \frac{-2}{n};\ B = -\frac{2m}{3n}$$

and initial conditions becomes $u(0) = \frac{3}{2m}$ $\quad u^{'}(0) = 0 \quad u^{''}(0) = \frac{3}{2mn}$. Apply Differential transformation to these initial conditions $U(0) = \frac{3}{2m}$; $U(1) = 0$; $U(2) = \frac{3}{4mn}$ initial conditions in Eq. (12), we get the differential transformation coefficients as

$$\begin{aligned} U(3) &= 0, & , U(4) &= \frac{1}{16mn^2} \\ U(5) &= 0, & , U(6) &= \frac{1}{48mn^3} \\ U(7) &= 0 & U(8) &= \frac{11}{5376mn^4} \end{aligned}$$

Using these values, Eq. (2) with $\omega = 1$ and $k = 1$ then $\eta = x + t$ then we have

$$u_4(x+t) = \frac{3}{2m} + \frac{3}{4mn}(x+t)^2 + \frac{1}{16mn^2}(x+t)^4 + \frac{1}{48mn^3}(x+t)^6 + \frac{11}{5376mn^4}(x+t)^8 \ldots \quad (16)$$

It is observed that solution $u(x, t)$ in Eq(16) is additive inverse of the solution of Eq. (15). The profile is represented as Fig. 6. The profile of Solution represented in Eq. (16) is inverted to the profile of Eq. (15), which is visible in Figs. 6 and 5. It is observed that the compacton-like wave is translated for phase 180 degree with the change in sign of parameter b, or it has the anti-campacton wave-like solution. This wave region becomes flat for the certain values of b, and the parameter a controls the wave amplitude.

4 Conclusion

In this paper, using the differential transformation method (DTM) method along with the wave-like transformation, we studied the numerical solution of nonlinear ZK-BBM equation. In numerical simulation, the finite series of DTM method is used and graphical profile for each case is elaborated. Behavior of the solutions is analyzed for every possible choice of parameters. The solution profile resembles for $a > 0, b > 0$ and $a < 0, b < 0$ or $a > 0, b > 0$ and $a < 0, b < 0$ shows that the numerical solution provides a pair of compacton wave-like solution of the differential equation, while other two cases are the copy of these solutions.

Author Contributions All persons who meet authorship criteria are listed as authors, and all authors certify that they have participated sufficiently in the work to take public responsibility for the content, including participation in the concept, design, analysis, writing, or revision of the manuscript. Furthermore, each author certifies that this material or similar material has not been and will not be submitted to or published in any other publication.

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Declarations

Conflicts of interest The authors declare that they have no conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

Informed consent Not applicable.

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